



An optimized XGBoost technique for accurate brain tumor detection using feature selection and image segmentation

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ABSTRACT

An abnormal multiplication of cells in the brain forms malignant and benign brain tumors. Malignant brain tumors are more prevalent than benign ones. Detecting a tumor's physical features may be tedious and time-consuming for medical experts due to the complexity of a tumor's structure and noise growth in Magnetic Resonance Imaging (MRI) data. Detecting and localizing tumors at an early stage in their growth is crucial. Complex structures and precise clinical identification using segmentation are considered the norm in medical imaging. As the quantity of images rises, the radiologist's manual tumor assessment might result in an incorrect conclusion. Due to the likelihood of human error, the evaluation and categorization of medical images require an automated method. This paper optimizes the eXtreme Gradient Boosting (XGBoost) approach through image processing and feature selection for the reliable identification of brain tumors. Images are improved using the Contrast-Limited Adaptive Histogram Equalization (CLAHE) method. The K-Means algorithm divides images into sections. This segmentation assists in pinpointing the specific area of interest. Particle Swarm Optimization (PSO) is used to choose characteristics. The data is classified by XGBoost, Naive Bayes, and the Iterative Dichotomiser 3 (ID3). According to the experimental results, the suggested PSO-XGBoost model achieved 97% accuracy, 97% specificity, 98% precision, and 98% recall.

1. Introduction

The term “brain tumor” refers to an abnormal growth caused by an excessive reproduction of brain cells. Brain tumors can be either cancerous or benign. Tumors in the brain are typically malignant rather than benign. Primary brain tumors consist of a group of cells with bloated nuclei that have formed in the brain. The expansion of the cell only occurs in the brain and nowhere else in the body. There are no malignant cells in a benign brain tumor. The tumor's development is not too far along, so surgical removal is still an option. When this lesion presses on a specific brain area, it will cause significant difficulties. The prognosis for survival will vary widely depending on where the tumor is located [1].

When cells in the brain multiply uncontrollably, a disease known as cancer may develop as a tumor. Uncontrolled cell growth and disruption of the body's regular cycle of cell death are the leading causes of many malignancies. It is possible, but also rather frequent, to have both primary and metastatic brain tumors. A benign tumor poses no danger to the healthy brain tissue around it and does not metastasize to other parts of the brain. When tumors in the brain develop fast and metastasize to other brain regions, we call them malignant. Benign tumors do not affect the surrounding tissue and

organs, whereas malignant ones (sometimes called cancerous ones) do [2].

When examined under a microscope, benign brain tumors can occasionally be shown to contain cells with a normal appearance. These tumors cannot spread to other brain parts because their distinct boundaries create a protective capsule.

Malignant brain tumors are highly uncommon and often localized to the brain, but there is a great danger that they will metastasize (spread to other body parts). However, these tumors are almost exclusively found in the brain. Regardless of how swiftly or slowly they progress, malignant tumors of this type always represent a severe threat to life since they can infiltrate and destroy nearby healthy cells. Malignant brain tumors are difficult to remove from the brain and spine without damaging healthy brain tissue [3,4]. This is because the tumors themselves lack distinct boundaries.

It may take time and effort for doctors to determine the tumor's physical features due to the complexity of tumor anatomy and the prevalence of noise in magnetic resonance imaging (MRI) data. Therefore, early detection and localization of tumors are crucial. In order to diagnose cancer early, a medical scan monitors and predicts the unchecked growth of potentially dangerous regions at different levels. Methods of categorization and demotion help achieve this goal. Cancer

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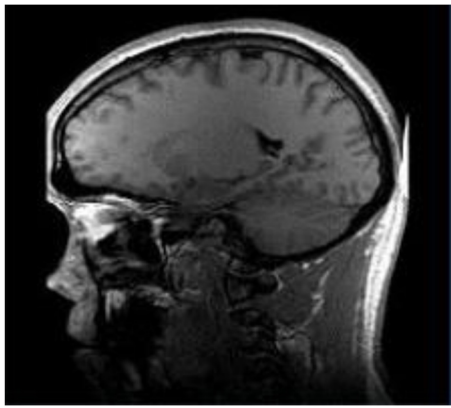


Fig. 1. MRI of Brain.

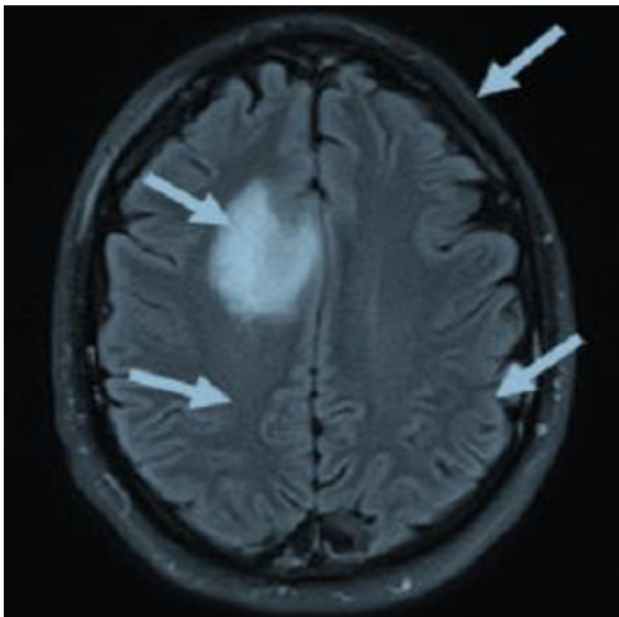


Fig. 2. MRI of Brain Tumor.

tissue detection in MRI scans requires substantial and time-consuming picture segmentation [5,6].

Accurate clinical identification using segmentation to reveal intricate structures is the gold standard in medical imaging. As more pictures are taken, the radiologist's manual assessment of the tumor increases the risk of an incorrect interpretation. Due to the scope of human error, medical image analysis and categorization needed to be automated. A radiologist's expertise is necessary to plan the numerous segmentation techniques effectively.

MRI-based temperature imaging is the only technology that can accurately measure temperature changes in real-time, which takes advantage of the temperature-sensitive water proton resonant frequency shift. Currently, this is the only viable option. Before being evaluated for use in clinical practice, this approach of guiding heat therapy has undergone extensive testing. Using this method, it is possible to accurately predict the amount of material that will be ablated, protect vital structures from damage, and lower the temperature below the injury threshold [7]. The validation experiments performed will be discussed in this paper. Clinical trials demonstrating the method's success when applied to human therapy will also be discussed—Figs. 1 and 2 show brain MRI images.

The use of magnetic resonance imaging (MRI), which is one of the most helpful modalities that are now in use, is one of the ways that advanced-stage brain tumors may be identified. When dealing with complex medical imaging, segmentation is an essential technique for determining the location of a questionable region. When it comes to the early diagnosis of brain tumors, automatic MRI tumor identification may provide a fresh perspective as well as a greater level of diagnostic accuracy.

Early detection and localization of cancers are crucial for their successful treatment. Complex structures and precise clinical identification via segmentation are regarded as the norm in medical imaging. There is a chance that the radiologist's manual assessment of the tumor, which is done as the number of images rises, will result in an incorrect conclusion. Given the potential for human error in the evaluation and classification of medical images, an automated method is necessary.

We came to the decision that it would be ideal to utilize this model while conducting our tests because XGBoost has the capability to tolerate null values, scale imbalanced data, learn from previous mistakes, and fine-tune hyperparameters. A method known as boosting is applied to succeeding ensembles in order to fix the errors that were introduced by prior models. This helps to ensure that the final product is as accurate as possible.

This article contains optimized XGBoost technique for accurate detection of brain tumor using feature selection and image segmentation. Images are enhanced using CLAHE technique. Images are segmented in parts by K Means algorithm. This segmentation helps in locating the exact region of interest. Features are selected by PSO particle swarm optimization technique. Classification is performed using XGBoost, Naïve Bayes and ID3 algorithm.

2. Literature survey

In order to identify brain tumors in MR images, Dina et al. [8] showed how to get better picture segmentation. This strategy was implemented. The Learning Vector Quantization (LVQ) model is achieving a 100% success rate, and a 79% reduction in computational time in image and data analysis. They looked into and spoke about linear and Gaussian filters, weighing the pros and cons of each. *Canny Edge Detection* is a method that has contributed to edge detection studies.

In order to expedite medical diagnosis, authors in [9] simultaneously maximized both accuracy and processing speed when applied to magnetic resonance (MR) images. In addition to enhancing the overall image quality, they have also decreased the amount of noise present. These researchers have studied many issues, such as the many kinds of brain tumors, image enhancement, morphological processing, and segmentation methods, and the differences and similarities between different intensity value ranges.

Rohini Paul Joseph and coworkers proposed using K-means clustering for brain MRI image segmentation [10]. After an MRI image of the brain has been segmented, a morphological process called structure-following-morphological-filtering is used to aid in tumor localization by preventing the creation of incorrectly grouped regions.

The approaches and techniques utilized for detecting brain tumors by MRI image segmentation were reviewed in depth by Mohamed Sabbih Hamoud Al-Tamimi and colleagues [11]. At the review's end, they included a brief commentary that might serve as a springboard for more studies into brain picture segmentation, thresholding, noise reduction, and tumor identification.

The confidence surface is a new idea Kishore R. Reddy and his team developed. Its goal is to facilitate segmentation with multi-parametric MR images as input [12]. The classifier's texture and intensity information was also trained using information from contrast-weighted images captured before and after T - I and T - II.

Analytical methods for identifying lesions and tumors in digital medical images were proposed by Phooi Yee Lau et al. [13]. This technique can be utilized to visualize unusual occurrences, such as lesions

or tumor cells included in image blocks. The suggested technique not only successfully identifies tumor blocks, but it also requires little in the way of computer resources and is conceptually essential.

Megha Soni et al. [14] showed how to replace qualitative visual analysis of picture data with quantitative processes to enhance visual discrimination between scene components. They achieved this by relying on the uniqueness of each picture's pixels to automate the detection of quantitative features. Because of this, quantitative methods could take the role of visual analysis. They have explained digital photo processing as a means to facilitate data collecting.

Hung et al. [15] proposed an artificial neural network with unsupervised learning capabilities; this network, dubbed "Self-Organizing Map" (SOM), was later developed. In addition, Fuzzy-Soft Learning Vector Quantization (FSLVQ) was created to enhance the SOM's adaptability to new information; for better performance of FSLVQ, the Suppressed-FSLVQ (or S-FSLVQ) was developed. The SFSLVQ method was superior to the other methods studied in a challenge requiring the segmentation of MR images of the brain. The FSLVQ draws on several different methodologies, including the generalized LVQ, the updated generalized LVQ, and the alternative LVQ, to name a few. Several more approaches can be taken. Numerical data generated by normal distributions were employed to compare and contrast the approaches that might be used. The algorithms' total performance was measured by considering their efficiency and accuracy. The proposed S-FSLVQ approach outperformed the state-of-the-art methods while reducing the computing effort needed. S-FSLVQ was proven more effective than competing learning algorithms at separating the tumor from normal brain tissue.

The use of neural networks based on unsupervised learning for the categorization of brain tumors was first proposed by Goswami and Bhaiya [16]. Picture preprocessing and tumor extraction from MRI scans are the first two steps in diagnosing a suspected brain tumor. The brain characteristic (ICA) was extracted using Independent Component Analysis. The self-organized map (SOM) is the gold standard for diagnosis in the third stage of determining whether or not a patient has a brain tumor. Segmentation of the brain pictures was performed using the K-Means method. The earlier experiment showed that unsupervised learning might effectively classify photos of brain tumors. The photographs were segmented to prove my point.

Brain tumors may now be detected automatically, according to a method developed and demonstrated by Ain et al. [17]. This strategy was demonstrated to be reliable. Multiple extraction steps were used in order to identify the brain tumor positively. The degree of background noise in brain MR images has been reduced. The attributes were retrieved once the noise-free brain pictures were acquired. After the feature was extracted, the various types of brain tumors were categorized accordingly. The brain tumors were classified using an Ensemble-based Support Vector Machine (SVM). The accuracy of a support vector machine (SVM) based technique to categorization was 99.1%. In order to remove the tumor from the MR images of the brain, the classification system employs methods that include many stages of segmentation. Once the skull was removed, for instance, cancerous brain tissue was isolated using Fuzzy C Means (FCM) clustering techniques for inclusion in the algorithm.

To locate the tumor precisely in the brain, researchers followed the foundational method proposed by Selvakumar et al. [18]. The type of the tumor was identified by analyzing magnetic resonance imaging (MRI) scans of the brain. Several patients who died due to the investigation into brain tumors received the incorrect diagnosis. The K-Means and Fuzzy C-Means algorithms were utilized to develop computational methods for tumor identification in the brain. In this study, three different automatic brain tumor segmentation algorithms were compared against manual brain tumor segmentation. The time needed to physically examine the brain tumors to determine their location and size was greatly reduced. The MR image was processed to remove the tumor, and the cancer stage was calculated automatically based on the density of the tumor's clusters. Their study found

that K-means clustering automatically removed tumors from MR brain pictures. Researchers chose to incorporate it within their suggested methodology alongside preprocessing techniques.

Brain tumor tissue should be automatically extracted from MR images, as Vijay et al. suggested [19]. Computer-aided approaches for identifying aberrant tissue growth are progressing because of the need to achieve the most significant feasible degree of accuracy. Modern, super-fast computers have made it possible to pinpoint the exact position and number of unwanted tissues. In this case, K-means clustering is employed as a segmentation technique. The tried-and-true Kmeans clustering technique was employed to identify the first cluster centers. The centroids might have been initialized in a way that produced empty or underperforming clusters.

Iterative convergence is a feature of the traditional K-Means approach for clustering data. However, it might lead to inaccurate results if the cluster centers are too similar. Vaishnavi et al. [20] proposed utilizing the Self Organizing Map (SOM) clustering approach to segment brain pictures. The histogram is normalized before the pictures are broken into segments so that characteristics may be retrieved. Principal Component Analysis (PCA) was used to determine the features, significantly improving the classifier's accuracy. Proximal Support Vector Machines (PSVM) were also developed as an alternative type of classifier. In comparison to SVM classifiers, these performed better. It has been demonstrated that the SVM classifier is a reliable classifier for labeling images.

Aslam et al.'s [21] thresholding methodology and the Sobel edge detection method were employed to locate the cerebral borders. Brain pictures were analyzed using a closed contour algorithm to segment the brain into its constituent parts. Brain tumors were effectively localized using the intensity data contained inside closed outlines. In medical imaging, it is common practice to observe intensity variation. Since of this, more than data on intensity alone is required since pictures near the margins are very faint and disordered.

A Deep Convolutional Symmetric Neural Network (DCSNN) was presented by Chen et al. [22] as a method for effectively segmenting the brain tumor. A glioma tumor, which represents a considerable risk of death, can only be treated through surgical removal, which is the only choice. Because of the use of MRI, the glioma was more easily accessible and more straightforward to therapeutically treat. It was essential for both the treatment and diagnosis planning processes that the glioma tumor be accurately segmented.

Random Forest (RF) and small kernels two-path convolutional neural network (SK-TPCNN) are two methods for autonomous brain tumor segmentation that were presented by Yang et al. [23]. The glioma diagnosis and treatment planning were significantly aided by the tumor segmentation performed using MRI. The great variety of glioma structures caused accuracy issues during the tumor segmentation process. For the purpose of carrying out effective tumor segmentation, the capability of optimization was paired with the capability of feature extraction.

A neural network-based end-to-end tumor segmentation strategy was developed by Li et al. [24], who made use of the image slices in both the testing and training phases of their research. This one worked better and was quicker than the patch-wise techniques, which were used in the previous ones. A brand new "up skip connection" has been incorporated into the design of the U-net in order to facilitate the addition of more "inception modules". Although the end-to-end segmentation method enhanced performance, the process was time-consuming because of the complexity of the brain tumors and the number of them.

Mlynarski et al. [25] created a CNN-based segmentation algorithm in order to improve the process of tumor segmentation training. By utilizing a hierarchical decision model that contained a large number of segmentation modules, it was able to circumvent the limitations imposed by the network designs. The primary purpose of this model was to improve the reliability and effectiveness of the CNN-based system. The multi-class segmentation that was developed by the method

that is generally accepted was combined with the voting strategy that was used for segmenting the structure of the tissue. It was able to attain greater overall performance by improving both the dice scores it received and the rate of precision at which it segmented the data. The memory needs of this model, on the other hand, as well as the processing time, skyrocketed.

Gupta et al. [26] reported a method for detecting gliomas in brain MRI scans. This method made use of both morphological and textual data. This methodology assisted radiologists in locating malignancies and provided doctors with helpful direction on how to accomplish the same task. In order to identify the glioma tumor, multiple classification levels were carried out utilizing the ensemble classifier and the texture fusion features that were based on the pulse sequences.

Sriramakrishnan et al. [27] developed a method for the automatic diagnosis of tumors by making use of a dataset comprising many sequences of MRI images. This methodology is comprised of three steps: identifying slices with tumors, extracting tumors, and segmenting the substructure of tumors. Initially, the image slices were analyzed with an SVM classifier and feature blocks in order to determine whether or not they contained neoplastic or non-neoplastic tissues. Second, it used an algorithm called the Fuzzy C-means Algorithm (FCM) to locate the part of the sliced images that was affected by the brain tumor.

Automatic glioma tumor segmentation was accomplished by Wang et al. [28] through the construction of the wide residual network and pyramid pool network (WRN-PPNet) model. The difficulty of tumor segmentation is contributed to by a number of factors, including tissue heterogeneity, fuzziness at the border between tumor and normal tissue, modality bias, and inhomogeneous gray in MRI. The majority of the approaches to tumor segmentation were laborious since they depended on a semi-automatic model, which made it challenging to localize the region affected by the tumor.

Deepak and Ameer provided a method for the classification of tumors that was based on deep learning in the paper [29]. Using MRI scans of the brain, it was able to successfully extract features from those scans. There was some consideration given to distinguishing pituitary tumors from those of the meninges and gliomas. We came up with the concept of a classifier so that we could put the collected image attributes into appropriate categories. The MRI database was utilized in a procedure known as cross-validation, and precise records of the categorizing process were maintained. Performance criteria such as precision, F-score, recall, and specificity were among those that were evaluated despite the limited amount of data available.

For the purpose of classifying various types of tumors, Swati et al. [30] created a deep convolutional neural network (CNN) classifier as well as a block-wise fine-tuning technique. When it comes to patients that have malignancies, clinical decision making and diagnosis are significantly reliant on accurate tumor classification achieved through MR imaging. Nevertheless, a fundamental barrier that must be overcome in order for MR images to be used for tumor classification is the semantic information gap that exists between the machine's detection of low-level information and the human's receiving of high-level data. This method proved to be fairly generalizable as there was a relatively little amount of preparation required for the 2D images. Unfortunately, this strategy did not work to boost performance using normal brain imaging.

Zhang et al. [31] came up with the concept of an attenuation residual U-Net for the segmentation of brain tumors. The design incorporates three primary categories of modules, which are referred to as contextual fusion, feature learning, and feature recovery respectively. The adaptive feature processing that takes place in the learning and retrieval module leads to better segmentation results, which in turn leads to an improvement in the tumor lesions. However, because of limitations in the computational source, information regarding the context of the tumor lesion parts is lost.

Gumaste and Bairagi [32] are the ones responsible for developing the hybrid feature extraction method for tumor identification in MRI

brain images. A grey level gap length matrix, run length encoding, a level size zone matrix, and a GLCM are utilized in order to create a partition that is used to split the provided photographs in half. This method can still be used for axial and coronal slices even if only half of the MRI needs to be scanned before texture characteristics can be extracted to reduce the amount of time required for processing.

Badza and Barjaktarovic [33] classified MRI images of brain malignancies by the application of CNN. A CNN architecture with twenty-two layers can classify a wide variety of malignancies, including gliomas, pituitary tumors, and meningiomas, to name just a few of them. Despite the fact that CNN has higher generalization capabilities, the performance validation of CNN is only available for T1-c MRI at this time. Cancer is one of the health problems that is most widely known about and experienced in today's society. According to the World Health Organization (WHO), over four lakh people all over the world are diagnosed with malignancies every single year [34]. Because of this, it is essential to detect and treat cancer at an earlier stage. In order for us to maintain a healthy lifestyle, the cells that make up our bodies must perpetually divide and multiply, with the number of cells in our bodies rising by the millions on a daily basis.

Deep neural networks, as opposed to traditional feature engineering, automatically generate a hierarchical data representation by extracting complicated hierarchies from raw data. This is done in contrast to traditional feature engineering. We offer a learning paradigm that incorporates a number of different representations so that students can hone their ability to abstract at greater levels. The model makes use of many levels of representation in order to make sense of the data. Each level of representation is used to extract a distinct collection of features from the data. The detection of cancers in photographs was investigated by Mansi Lather and colleagues [35] using a number of different machine learning techniques that were based on soft computing.

As part of their Deep Learning approach to brain tumor classification and segmentation, the authors of [36] describe a multi-path-way Convolution Neural Network architecture for tumor segmentation. The three branches of the model are integrated using a single convolution layer and max-pooling, and each of the branches contains two max-pooling convolution layers of variable sizes.

Gupta et al. [37] provides a comprehensive description of the image registration survey. Within the scope of this investigation, we presented a number of methods for the anatomical and functional registration of medical photographs. This is a study of the image registration processes that can be processed between anatomical and functional medical images. It includes references to a variety of different types of medical imaging, such as MRI, CT, PET, FLAIR, and so on. An analysis of a number of different methods for image registration revealed that the supervised learning model provided the most significant improvement to the inter-image feature prediction.

Because of this, the deep learning strategy was selected in preference to the weekly supervised learning model proposed by Blendowski et al. [38]. When compared to its fully supervised cousin, a partially or only partially supervised learning model requires a smaller training dataset in order to correctly identify the primary characteristics of a picture. This can further reduce the complexity of the space requirements during the training process for classification models.

Hassan Habib and colleagues [39] investigated the growth of aberrant cells in the human brain by employing a strategy that involved the hybridization of image segmentation and a classification model, in addition to the procedure for the extraction of feature information. During the image segmentation procedure, both the threshold segmentation approach and the watershed algorithm were utilized in order to hone in on the picture cluster that contained the tumor. In the hybrid model of image feature extraction, many image representation algorithms such as MSER, FAST, and Harlick, among others, are utilized. After then, a variety of classifiers were utilized in order to produce predictions regarding the labels of the MRIs. The classification model's foundation is a texture learning model, which is used to do analysis on the image's defining characteristics.

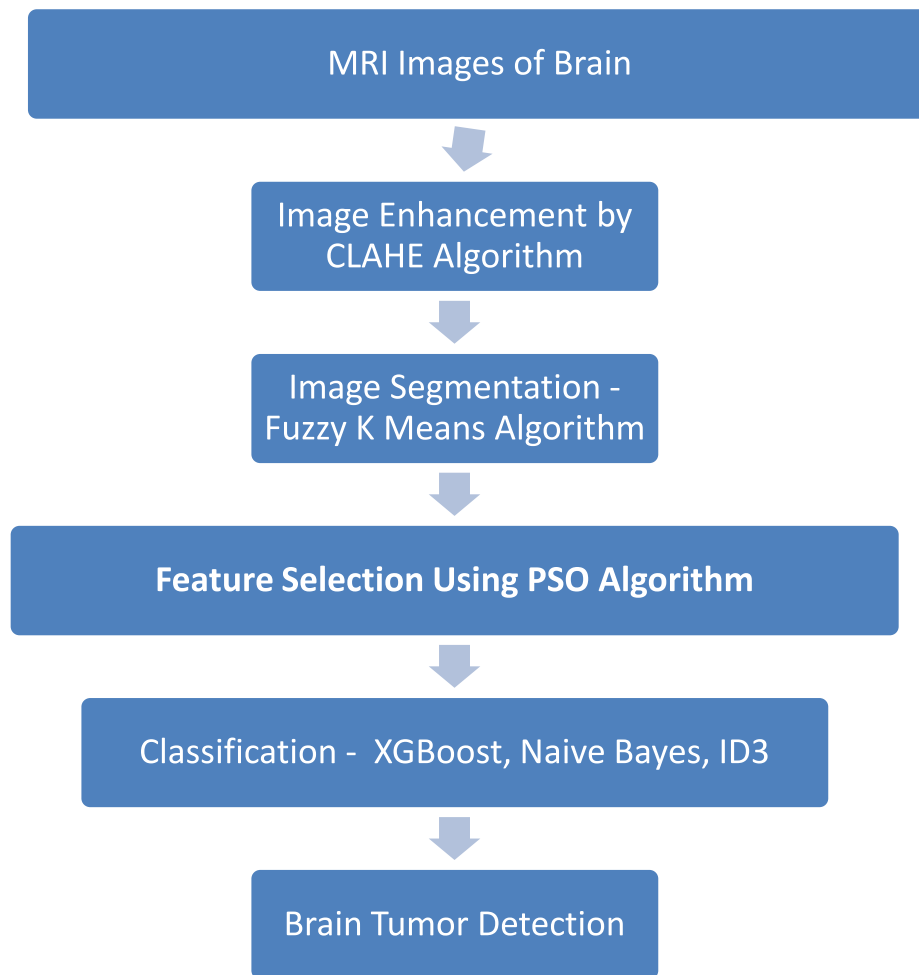


Fig. 3. Optimized XGBoost technique for accurate detection of brain tumor using feature selection and image segmentation.

3. Methodology

This section contains optimized XGBoost technique for accurate detection of brain tumor using feature selection and image segmentation. Images are enhanced using CLAHE technique. Images are segmented in parts by K Means algorithm. This segmentation helps in locating the exact region of interest. Features are selected by PSO particle swarm optimization technique. Classification is performed using XGBoost, Naive Bayes and ID3 algorithm (see Fig. 3).

To make the photos seem better, contrast-limited adaptive histogram equalization (also known as CLAHE) was used [40]. It is possible that pre-processing methods like CLAHE might make skin lesions easier to see and see more clearly. CLAHE is able to provide a contrast range that is both small and consistent because to its use of bilinear interpolation and bilinear thresholding. The collection includes pictures in a range of different dimensions. There will be a difference in the feature values that are extracted from photographs of different sizes. In this technique, the image's pixel dimensions were scaled down to 768 pixels by 576 pixels. In order to save all of the information from the original photograph, a resolution of 768 by 576 pixels was perfect. The photographs are now eligible to go on to the process of segmentation.

Image segmentation is accomplished with the help of the K-means algorithm [41]. For the purpose of classifying or partitioning your data into k separate categories, a method called as K-means may be used. k is a positive integer; it cannot be negative. When data that are similar to one another are grouped together, a cluster is created because the sum of squares of the distances between the data points and their respective cluster centroids is decreased. In a nutshell, the data are classified using

k-means clustering as the method of choice. The K-means clustering method is an easy-to-understand procedure. There are k clusters, and we are going to assume that the number k serves as the centroid of all of these clusters. The initial centroids may be picked at random, or they can be selected from the first k items. Either way, they must come from the same set. The K-means algorithm will then go through the three steps listed below until it reaches convergence:

Iterate until stable (= no object move group):

1. Determine the centroid coordinate
2. Determine the distance of each object to the centroids
3. Group the object based on minimum distance (find the closest centroid)

The pattern of behavior seen in flocks of birds and schools of fish will serve as the inspiration for the PSO, which is a stochastic method. A search space contains a number of particles, each of which has its own position and velocity, both of which are calculable. The movement and acceleration of the particle are still being regulated, and they are being directed to the position of a particle that is doing well, as well as to the place that was previously fantastic for all other particles [42]. Because the velocities and locations of particles change throughout the course of time, a set of rules has been constructed to characterize this process. These laws may be found here.

A representation of the sequence model known as XGBoost [43]. Because of XGBoost's capacity to accept null values, scale imbalanced data, learn from previous mistakes, and fine-tune hyperparameters, our team came to the conclusion that it would be best to use this model while conducting our experiments. A technique known as boosting is used to successive ensembles in order to correct the faults that

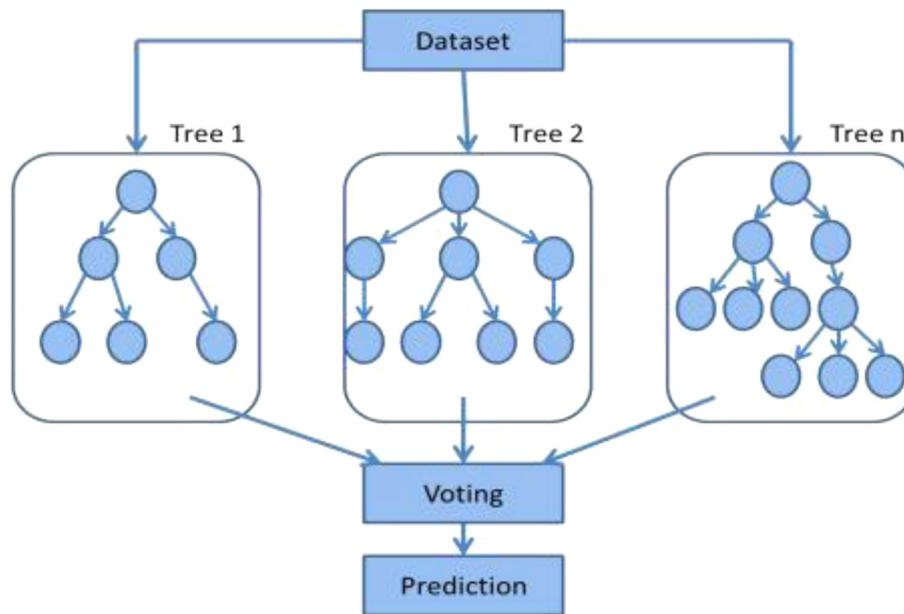


Fig. 4. Extreme Gradient Boosting.

were introduced by earlier models. It has been shown that the use of XGBoost may enhance weak learners' performance in classification and regression tasks. If the previous prediction value for the data that was supplied as input is taken into consideration, then XGBoost may generate new trees in order to optimize the prediction improvements it produces. It is shown below in Fig. 4

As a direct consequence of these developments, a new specialized field known as Bayesian inferences has emerged very lately. It is not necessary to provide a clear estimate of the iterative parameters in order to create naïve Bayesian models. It is possible to calculate the posterior probability of $P(c|x)$ by adding the likelihoods of $P(c, x)$ and $P(x|c)$, which is a process that may be carried out with the assistance of the Bayes theorem. It is feasible to calculate the posterior probability by first developing a frequency table and then conducting a number of statistical tests on the data. The naïve Bayesian approaches are used to the frequency tables in order to do the calculations necessary to determine the probability for each group included in the dataset. After taking into account all of the information that is readily accessible, the classification that has the highest posterior probability is selected [44].

The ID3 technique of Decision Trees [45] is used to build a tree that considers many criteria and possible values for the interest classes. In any programming language, statements can be stacked in an if-else tree structure. ID3 utilizes entropy and information gain to conduct statistical analysis on training data and summarize the results. Because of its greedy nature, this algorithm does not offer backtracking as a problem resolution. Standard ID3 does not employ any optimization or pruning strategies regarding tree size. The degree of similarity between data points may be measured by computing their entropy. To have a shared worldview is what is meant by "homogeneity". Its entropy is one if and only if the values it stores statistically represent the whole population. This condition can only be met by calculating entropy at the attribute level or combining two attributes.

4. Result analysis

For experimental work, total 250 images are selected at random [46]. 150 images are used for training of model and remaining 100 images are used for the testing of the model. Images are enhanced using CLAHE technique. Images are segmented in parts by K Means algorithm. This segmentation helps in locating the exact region of interest. Features are

selected by PSO particle swarm optimization technique. Classification is performed using XGBoost, Naïve Bayes and ID3 algorithm. Results of experimental work are shown in Figs. 5, 6, 7 and 8. Accuracy of XGBoost method is better for the detection of Brain tumor.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$\text{Sensitivity} = TP / (TP + FN)$$

$$\text{Specificity} = TN / (TN + FP)$$

$$\text{Precision} = TP / (TP + FP)$$

$$\text{Recall} = TP / (TP + FN)$$

Where

TP = True Positive

TN = True Negative

FP = False Positive

FN = False Negative

5. Conclusion

Early detection and localization of cancers are crucial for their successful treatment. Complex structures and precise clinical identification via segmentation are regarded as the norm in medical imaging. There is a chance that the radiologist's manual assessment of the tumor, which is done as the number of images rises, will result in an incorrect conclusion. Given the potential for human error in the evaluation and classification of medical images, an automated method is necessary. In this study, feature selection and picture segmentation are used to demonstrate how the XGBoost approach can be improved for precise brain tumor diagnosis. Using the CLAHE method will improve image quality. The K Means method divides up images to generate sections inside them. We were able to focus on the particular region of interest with the help of this segmentation. When choosing attributes, the particle swarm optimization (PSO) method is used. For data classification, XGBoost, Nave Bayes, and the ID3 algorithm are used. XGBoost performs better in detecting brain tumors. According to the experimental results, the suggested PSO-XGBoost model achieved 97% accuracy, 97% sensitivity, 98% precision, and 98% recall.

In light of the fact that a prompt and correct diagnosis of brain malignancies devoid of latency is required, our future study will focus on the development of alternative powerful hybrid techniques for brain

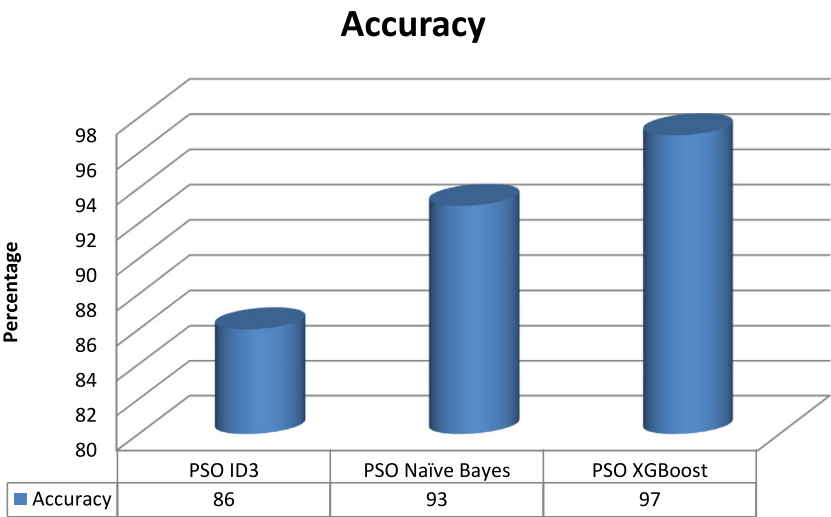


Fig. 5. Accuracy of Classifiers for Brain Tumor Detection.

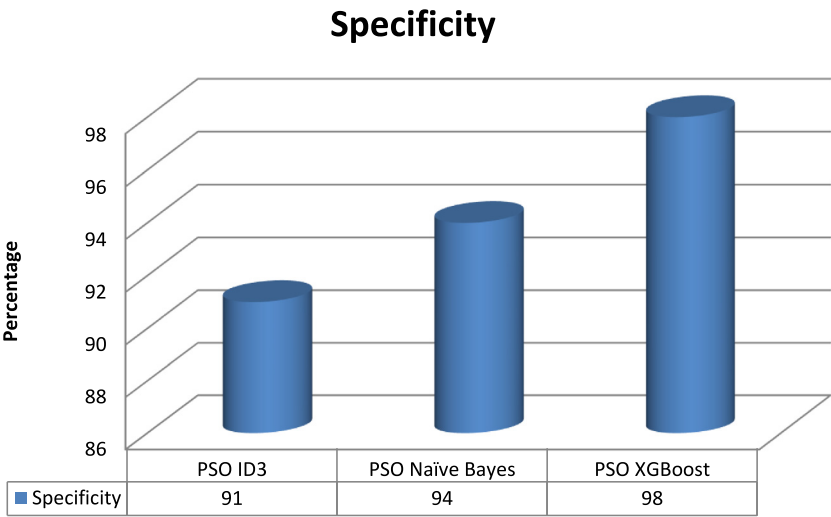


Fig. 6. Specificity of Classifiers for Brain Tumor Detection.

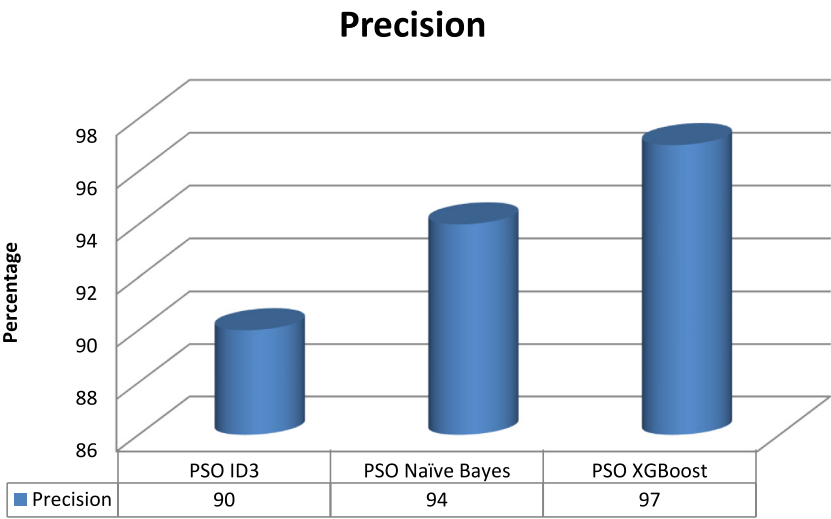


Fig. 7. Precision of Classifiers for Brain Tumor Detection.

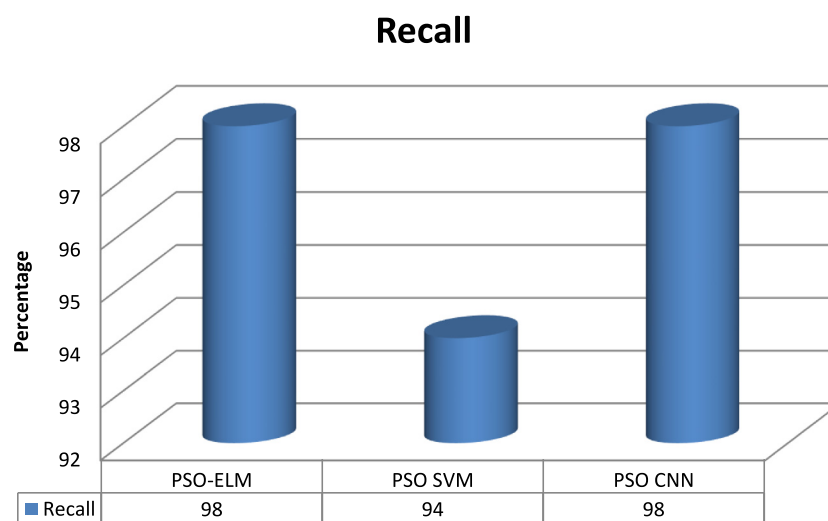


Fig. 8. Recall of Classifiers for Brain Tumor Detection.

tumor classification. These strategies will need less time to carry out and will provide a higher level of simplicity. In addition, the recommended procedures may be used in conjunction with MRI and CT scans to assist in the diagnosis of a variety of cancers.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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